

Corporate Valuation Methods

Session 2

Forecasting & Valuation Inputs

From historical economics to a defensible forward-looking valuation model



Session 1 → Session 2

SESSION 1 — DIAGNOSE

Understand the past

- Normalized EBIT
- NOPAT
- Invested capital
- ROIC
- Reinvestment
- Historical FCFF
- Historical FCFE



Analyze and normalize historical performance.

From
evidence
to
expectations

SESSION 2 — FORECAST

Build the future outlook

- Revenue drivers
- Margins
- Future ROIC
- Reinvestment
- Firm lifecycle
- Risk inputs & WACC



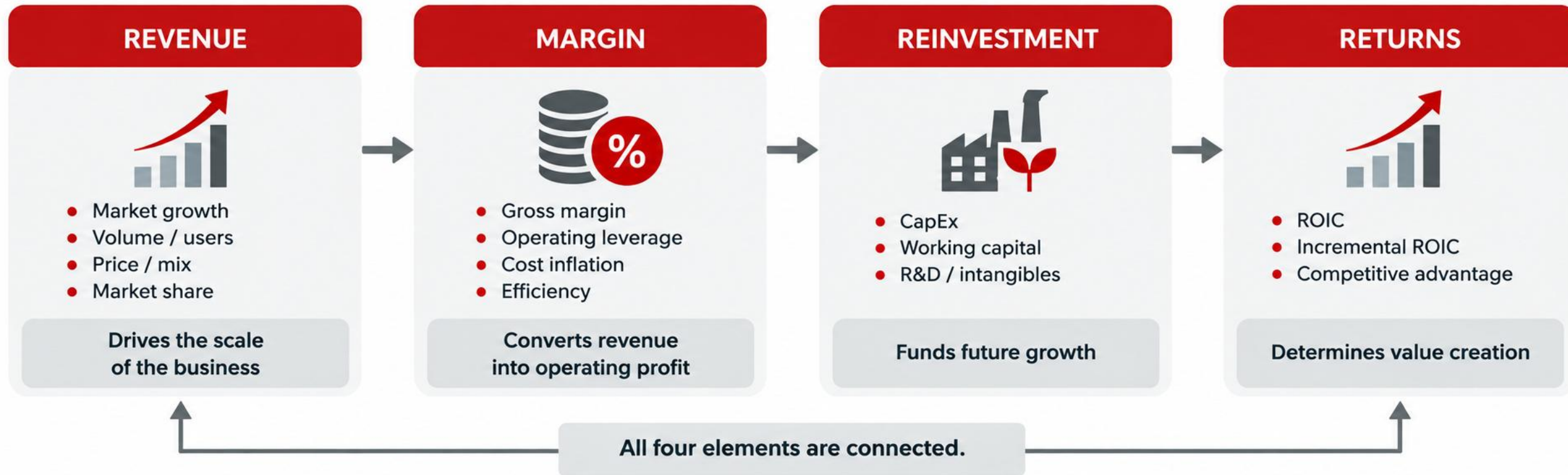
Build a consistent, forward-looking forecast.

Historical numbers are evidence — not the forecast.

The Real-Company Forecasting Workflow



A Forecast Is an Economic Story Expressed in Numbers



Every major forecast number should have a business explanation.

How Long Should the Explicit Forecast Be?

5 YEARS MAY WORK

For mature, stable businesses

- Mature business
- Stable margins
- Stable ROIC
- Limited structural change



7-10+ YEARS MAY BE NEEDED

For high-growth or transforming businesses

- High growth
- Margins still evolving
- ROIC far from mature level
- Major capacity build-out



KEY TEST

Look at the terminal year

- By the terminal year, is the company economically close enough to a stable state?



Year 1
Forecast starts

Terminal Year
Stable state

Forecast horizon is an economic choice, not a spreadsheet convention.

Revenue Forecasting — Start with Drivers

TOP-DOWN LOGIC

$$\text{Revenue} = \text{Market Size} \times \text{Market Share}$$



Market Size
(industry growth)



Market Share
(company position)



Revenue
(growth over time)

PRODUCT / INDUSTRIAL

$$\text{Revenue} = \text{Volume} \times \text{Price}$$



Volume
(units)



Price
(per unit)

Drivers: unit demand, pricing,
product mix.

SUBSCRIPTION

$$\text{Revenue} = \text{Customers} \times \text{ARPU}$$



Customers
(subscribers)



ARPU
(revenue per user)

Drivers: customer growth, retention,
pricing / upsell.

RETAIL

$$\text{Revenue} = \text{Stores} \times \text{Sales per Store}$$



Stores
(number of stores)



Sales per Store
(revenue per store)

Drivers: store expansion, same-store
sales growth, basket size.

Choose the driver structure that matches how the company actually makes money.

Forecast Segments Before the Group Total :

MATURE SEGMENT

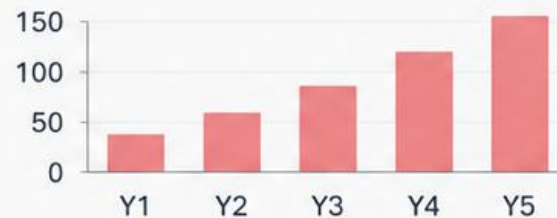
- Large base
- Low category growth
- Stable share
- Pricing $\approx$ inflation



Stable, steady growth

GROWTH SEGMENT

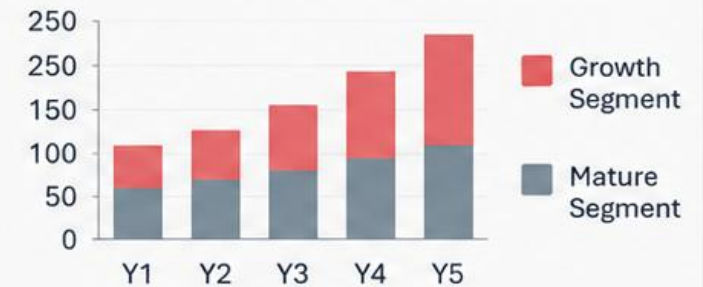
- Smaller base
- Faster category growth
- Potential share gains
- Pricing $>$ inflation



Higher growth, scaling up

GROUP FORECAST

- Sum segment forecasts, then challenge the consolidated result.



Consolidated view — but check the underlying drivers.

A consolidated 5% forecast may hide very different economics underneath.

Historical Growth Is a Starting Point — Not an Answer

LOOK BACK

What has happened?

- 3–5 year CAGR
- Organic vs acquired growth
- Volume vs price
- FX / one-offs

Historical Revenue



LOOK FORWARD

What will drive future growth?

- Industry outlook
- Capacity
- Market share
- Pricing power
- Guidance



Industry growth



Capacity



Market share



Pricing power

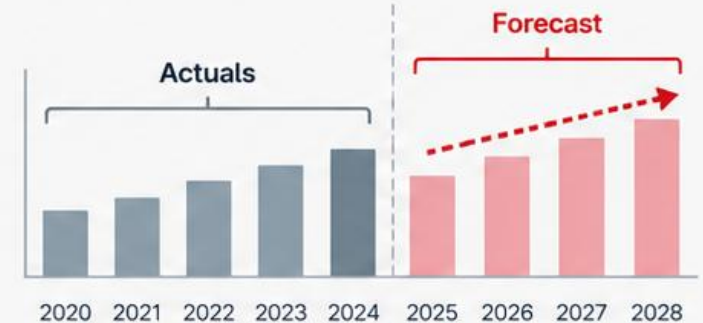


Guidance

RECONCILE

Should the future be similar or different?

- Why should the future be similar to — or different from — the past?



Never use historical CAGR only because it is the historical CAGR.

Worked Example — Build Revenue from Drivers

MARKET

How fast is the category growing?



Category growth
(Industry / total market)

Year	Category growth
2027	4.0%
2028	3.8%
2029	3.5%

Based on industry reports
and market research.

SHARE DEVELOPMENT

How much share can we gain?



**Expected contribution
from market share**
(Company competitive position)

Year	Share contribution
2027	+0.8%
2028	+0.6%
2029	+0.3%

Driven by better products,
distribution, capacity, etc.

IMPLIED REVENUE GROWTH

Market growth + share development



**Resulting company
revenue growth**

Year	Market growth	Share contribution	Implied revenue growth
2027	4.0%	+0.8%	4.8%
2028	3.8%	+0.6%	4.4%
2029	3.5%	+0.3%	3.8%

The forecast now has an economic source:
market growth + share development.

The forecast now has an economic source: market growth + share development.

Project Task — Build Your Revenue Assumption Table

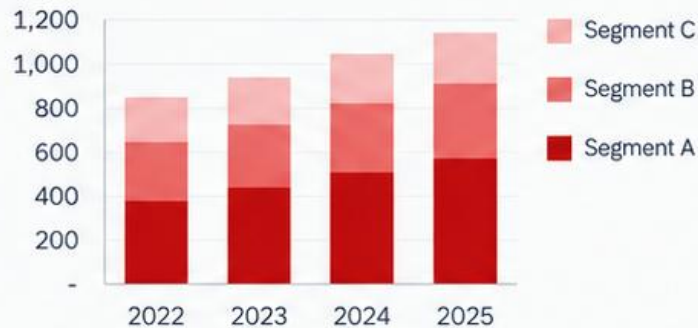
HISTORICAL

What has happened?

- Revenue by segment
- Organic growth
- Volume / price where available

Segment Revenue (historical)

€m



Understand the starting point and segment dynamics.

EVIDENCE

What supports your view?

- Company guidance
- Industry outlook
- Capacity / stores
- Market share



Company guidance



Industry outlook



Capacity / stores



Market share



Pricing trends



Competitive position

Use multiple sources to build defensible assumptions.

FORECAST

What do you expect?

- Driver-based growth
- Segment mix
- Convergence toward maturity

Revenue Forecast (example)

€m

Segment	2025	2026	2027	2028	2029	CAGR 25–29
Segment A	500	530	561	590	620	5.5%
Segment B	300	330	362	395	425	9.2%
Segment C	200	212	225	236	245	5.2%
Total	1,000	1,072	1,148	1,221	1,290	6.6%

Turn your analysis into a clear, driver-based forecast.

For every forecast year, be able to answer: what changed, and why?

Operating Margin Forecast — What Drives the Change?

PRICING / MIX

Can we charge more, or sell a different mix?

- Premiumization
- Price realization
- Product mix

Average Selling Price (€)



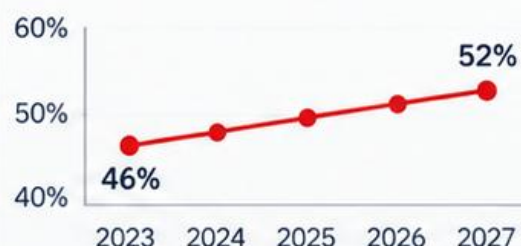
Higher prices or a richer product mix can lift margins.

GROSS MARGIN

What happens to input costs and productivity?

- Input costs (commodities, labor)
- FX
- Supply chain
- Productivity / efficiencies

Gross margin (%)



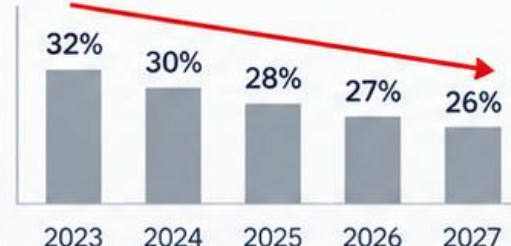
Lower input costs and higher productivity can expand gross margins.

OPERATING LEVERAGE

How does growth affect fixed costs?

- Fixed-cost absorption
- Scale benefits
- Cost discipline

Operating costs (% of revenue)



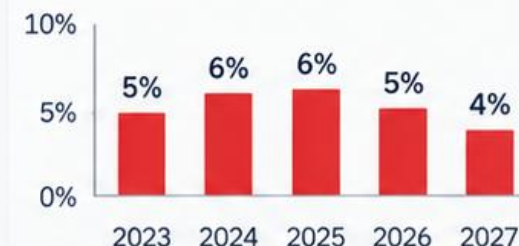
As revenue grows, fixed costs are spread over a larger base.

INVESTMENT

Are we investing for growth?

- Marketing
- R&D
- Digital
- New capacity

Investment (% of revenue)



Higher investment can weigh on near-term margins, but support future growth.

“+50 bps per year” is not a mechanism.

Worked Example — Build a Margin Bridge

2027E OPERATING MARGIN

$$12.0\% + 0.4\% + 0.3\% + 0.5\% - 0.4\% = 12.8\%$$



Build the bridge first; then decide whether the resulting margin is sustainable.

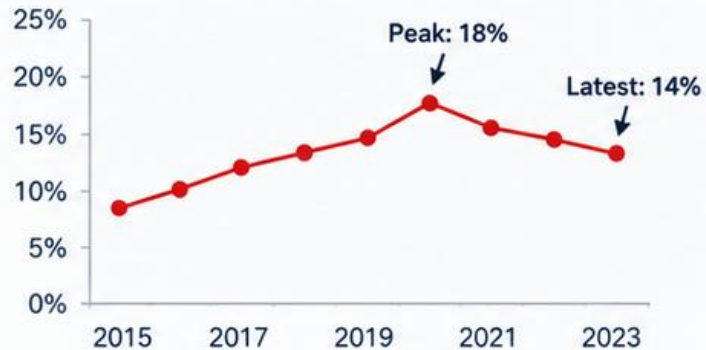
Margins Cannot Expand Forever

COMPANY HISTORY

Where have margins been through the cycle?

- Look at a full cycle (e.g., 5–10 years)
- Identify peak, trough and average
- Understand what drove past changes

Historical operating margin (%)



History shows the range of margins achievable in practice.

PEERS

What margins do comparable firms earn — and why?

- Compare with similar companies
- Adjust for business mix and scale
- Understand structural differences
- Look at peer margin ranges, not just the top

Peer operating margins (%)

Company	Operating margin
Peer 1	18%
Peer 2	15%
Peer 3	12%
Peer 4	10%
Peer average	14%

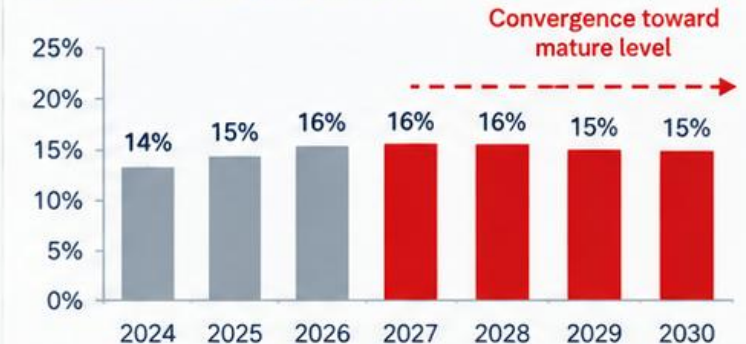
Your forecast should be consistent with a defensible peer comparison.

ECONOMICS

Does the company have pricing power, scale or structural cost advantages?

- Sustainable competitive advantages?
- Scope for further scale benefits?
- Any cost headwinds (e.g., inflation, labor)?
- Regulation, competition, technology change?

Illustrative margin path (%)



Margins may improve, but they rarely expand indefinitely.

The mature margin should reflect a sustainable competitive position.

Forecast ROIC — Do Not Leave It as a Residual

RECALL FROM SESSION 1

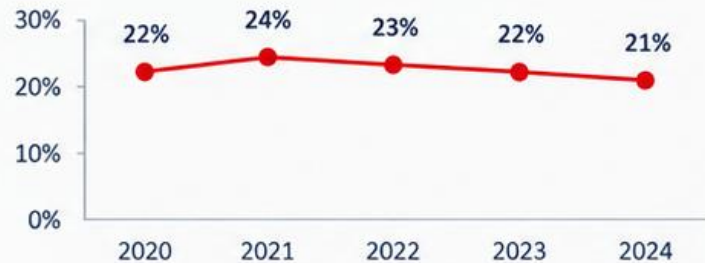
$$\text{ROIC} = \text{NOPAT} \div \text{Invested Capital}$$

HIGH & PERSISTENT

Structural advantages support high returns.

- Strong moat (brand, network effects)
- Low cost position
- High switching costs

ROIC (%)



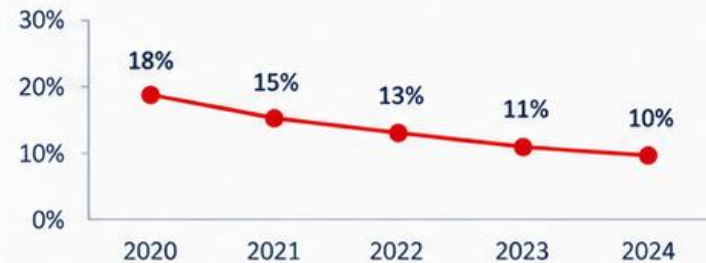
High ROIC can persist when the competitive advantage is durable.

CONVERGING

Returns tend to move toward the industry level.

- More competition over time
- Maturity of the market
- Imitation and rising capital needs

ROIC (%)



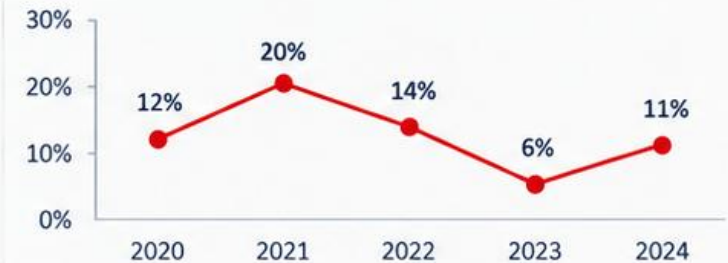
As the business matures, ROIC often converges toward the cost of capital.

CYCLICAL

Returns move with the cycle.

- Sensitive to utilization
- Commodity prices or input costs
- Upside and downside through the cycle

ROIC (%)



Forecast ROIC through the cycle, not just at the peak.

Forecast the economic story of ROIC — not just NOPAT.

Make Growth and Reinvestment Consistent

FUNDAMENTAL CONSISTENCY CHECK

$$\text{Reinvestment Rate} \approx \text{Growth} \div \text{ROIC}$$

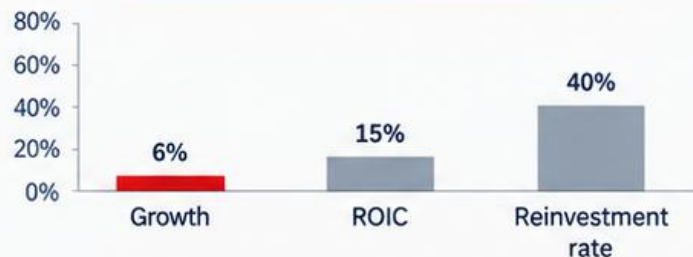
HIGH ROIC

Less capital needed to grow.

- 6% growth / 15% ROIC → 40% reinvestment rate.
- Can grow meaningfully with modest investment.
- Often seen in asset-light or high-ROIC businesses.

Illustrative example (High ROIC)

6% growth / 15% ROIC = 40% reinvestment



High ROIC allows growth with less capital.

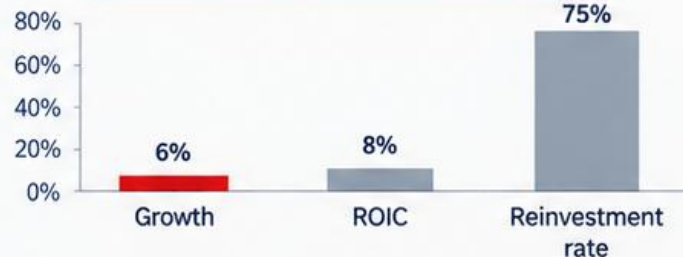
LOW ROIC

Much more capital needed to grow.

- 6% growth / 8% ROIC → 75% reinvestment rate.
- Growth is more capital intensive.
- May be harder to sustain high growth.

Illustrative example (Low ROIC)

6% growth / 8% ROIC = 75% reinvestment



Low ROIC requires high reinvestment to grow.

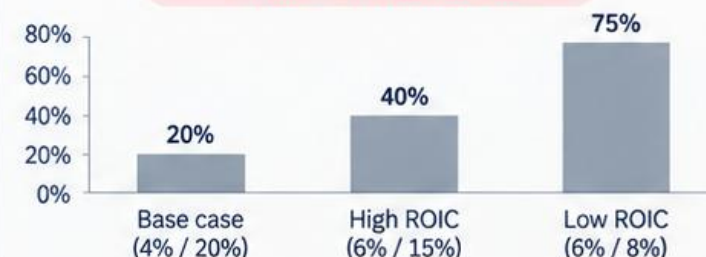
MODEL CHECK

Is the implied reinvestment reasonable?

- Strong growth with almost no reinvestment should trigger a question.
- Check against industry norms and company history.
- Ensure growth, reinvestment and ROIC are mutually consistent.

Illustrative check

Implied reinvestment rate



Check that implied reinvestment is plausible for the business.

Growth is not a free input.

Build vs Buy — Two Routes to Growth

Organic investment and acquisitions can partly substitute for one another.

BUILD

- CapEx, R&D, new stores, factories, technology, distribution and working capital.

BUY

- Acquire another company, product portfolio, technology, customer base or capacity.

FORECAST QUESTION

- How much of historical growth was organic?
- How much came from acquisitions?
- Is that acquisition strategy expected to continue?

Do not forecast acquired historical growth as if it were organic growth.

Capitalizing R&D — Why It Matters

KEY IDEA

R&D is often an investment, not just a period expense.
Capitalizing R&D makes EBIT, invested capital and ROIC more economically meaningful.



ACCOUNTING VIEW

Expense immediately

- Reported EBIT is reduced by current R&D.
- No R&D asset appears on the balance sheet.
- Profitability may look lower in high-R&D years.



ECONOMIC VIEW

Investment in future benefits

- R&D may create products, patents, software or know-how.
- Benefits usually arrive over several years.
- Treat it like an intangible capital investment.



VALUATION VIEW

Make performance comparable

- Add back current R&D to EBIT.
- Replace it with amortization of past R&D.
- Include the R&D asset in invested capital.

Capitalizing R&D does not create cash; it changes classification from operating cost to investment.

Capitalizing R&D — How to Do It

R&D ASSET ROLL-FORWARD

$$\text{R\&D Asset} = \text{Prior R\&D Asset} + \text{Current R\&D} - \text{R\&D Amortization}$$

Use a useful life assumption, such as 3–5 years depending on the industry.

STEP 1 Choose a useful life



- Software or tech: often shorter life.
- Pharma or long-cycle innovation: often longer life.
- Keep the assumption consistent with the business model.

STEP 2 Build the R&D asset



- Collect current and past R&D spending.
- Amortize each year's R&D over the chosen life.
- Sum the unamortized balance.

STEP 3 Adjust operating profit



- Remove current R&D expense from EBIT.
- Deduct R&D amortization instead.
- Tax-effect the adjusted EBIT.

Adjusted EBIT = Reported EBIT + Current R&D – R&D Amortization

Capitalized R&D — Incorporate It in FCFF

STANDARD FCFF

$$\text{FCFF} = \text{NOPAT} + \text{D\&A} - \text{CapEx} - \Delta\text{NWC}$$

When R&D is expensed, it is already included in reported EBIT and NOPAT.

FCFF AFTER CAPITALIZING R&D

$$\text{FCFF} = \text{Adjusted NOPAT} + \text{D\&A} + \text{R\&D Amortization} - \text{CapEx} - \text{Current R\&D} - \Delta\text{NWC}$$

Current R&D is an investment outflow; R&D amortization is a non-cash add-back.

ADD BACK Non-cash charges



- D&A on physical assets.
- Amortization of the R&D asset.

SUBTRACT Cash investments



- CapEx and change in NWC.
- Current R&D investment.

CHECK Economic consistency



- ROIC uses adjusted NOPAT.
- Invested capital includes R&D asset.

The cash-flow mechanics should reconcile; the analytical benefit is cleaner ROIC and reinvestment.

Worked Example — Capitalizing R&D

Assume current R&D = €100m, R&D amortization = €60m, tax rate = 25%

EBIT ADJUSTMENT

Replace expense with amortization



- Reported EBIT: €300m
- + Current R&D: €100m
- – R&D amortization: €60m
- = Adjusted EBIT: €340m

NOPAT IMPACT

Tax-effect the adjustment



- EBIT increase: €40m
- Tax rate: 25%
- NOPAT increase: €30m
- Adjusted NOPAT is higher by €30m

FCFF IMPACT

No free cash-flow magic



- Higher adjusted NOPAT: €30m
- + R&D amortization add-back: €60m
- – Current R&D investment: €100m
- = Net after-tax effect: about –€10m

The main benefit is analytical: better EBIT, invested capital, ROIC and reinvestment interpretation.

Working Capital — Small Assumption, Large Cash-Flow Effect

$$\text{Operating NWC} = \text{Receivables} + \text{Inventory} - \text{Operating Payables}$$



FORECAST

NWC / sales or operating days.



WATCH FOR

Factoring, supplier stretching, seasonality, acquisitions.



CHECK

Does growth require more working capital — or is negative NWC structural?

Do not assume zero ΔNWC simply because the historical line is small.

CapEx — Maintenance vs Growth



MAINTENANCE CAPEX

Required to sustain current operating capacity.

- Is D&A a reasonable proxy?
- Is the asset base old?
- Has CapEx been deferred?



GROWTH CAPEX

Supports additional capacity or new products.

- What projects are planned?
- When do they generate revenue?
- What return should they earn?

Low CapEx can temporarily boost FCFF while weakening the future operating base.

Do Not Hide Uncertainty in One Base Case



BEAR

- Lower market growth
- Less pricing power
- Slower margin improvement
- Lower ROIC



BASE

- Most defensible central operating case.



BULL

- Stronger share gains
- Better mix
- Operating leverage
- Higher ROIC

Scenarios should change economic drivers — not simply $\pm 1\%$ on every line.

Forecast Toward a Credible Mature State



GROWTH

Moves toward sustainable long-run nominal growth.



MARGIN

Approaches a mature competitive level.



ROIC

Converges toward a sustainable return.



REINVESTMENT

Becomes consistent with mature growth and ROIC.

The terminal year should look like a mature company — not Year 5 of an unfinished growth story.

Part II — From Operating Forecast to Discount Rate



FORECAST

What cash flows do we expect?

Revenue → margin → NOPAT → reinvestment → FCFF



DISCOUNT RATE

How risky are those cash flows?

Risk-free rate → risk premium → beta → debt cost → WACC

For a real company, every WACC input needs a source and a rationale.

WACC — The Applied Input Map

$$\text{WACC} = E/(D+E) \times K_e + D/(D+E) \times K_d \times (1-T)$$



COST OF EQUITY

- Risk-free rate
- Beta
- ERP
- Country risk



COST OF DEBT

- Benchmark rate
- Credit spread
- Tax rate



WEIGHTS

- Market-value equity
- Debt
- Target structure

The formula is easy. The valuation work is choosing the inputs.

Risk-Free Rate — Match Currency, Not Headquarters



PRINCIPLE

Use a default-free rate in the same currency as the cash flows.



EXAMPLE

EUR cash flows → euro risk-free benchmark.
USD cash flows → US-dollar benchmark.



DO NOT

Use a local government yield only because the company is headquartered there.

Currency drives the risk-free rate; operating geography affects country risk.

Country Risk — Exposure Matters More Than Incorporation

$$\text{Company CRP} \approx \sum (\text{Operating Exposure by Country} \times \text{Country Risk Premium})$$



MULTINATIONAL

A European-listed company may have substantial emerging-market exposure.



DATA

Revenue geography is a starting point; assets and production can also matter.



LOGIC

Country risk follows the cash flows.

Headquarters alone does not determine operating country risk.

Leveraged versus Unleveraged Betas

UNLEVERAGED BETA (β_u)

- In the absence of debt, the beta is called the unleveraged β_u .
- It is impacted by the firm's operating leverage and the cyclical nature of the industry in which the firm competes.

LEVERAGED BETA (β_l)

- In the presence of debt, the beta is called the leveraged β_l .
- It is impacted by the firm's operating leverage, industry cyclical nature, and financial leverage.

KEY ASSUMPTION

- If a firm's shareholders bear all the risk of operating and financial leverage and interest is tax deductible, leveraged and unleveraged betas can be calculated as follows:

$$\beta_l = \beta_u (1 + (1 - t) (D/E)) \quad \text{and} \quad \beta_u = \beta_l / (1 + (1 - t) (D/E))$$

where t , D , and E are the tax rate, debt and equity, respectively.

IMPLICATIONS

- Increasing D/E raises the firm's breakeven and increases shareholder risk that the firm will be unable to generate future cash flows sufficient to pay their minimum required returns.
- Tax deductibility of interest reduces shareholder risk by increasing after-tax cash available for shareholders.

Cost of Equity — Build It Transparently

$$K_e = \text{Risk-Free Rate} + \beta \times \text{ERP} + \text{Country-Risk Adjustment}$$

DOCUMENT

- Valuation date
- Currency
- Rf source

DEFEND

- Peer set
- Unlever/relever assumptions

EXPLAIN

- ERP source
- Country-risk treatment

A reviewer should be able to reproduce your cost of equity.

Cost of Debt — Use Current Credit Risk

OBSERVABLE Use market information

- Bond yield
- Recent debt issue
- Credit spread

SYNTHETIC Derive from credit quality

- Interest coverage / credit quality → default spread → $R_f + \text{spread}$.

AVOID Don't rely on outdated proxies

- Historical interest expense / average book debt without checking current relevance.

WACC is forward-looking; cost of debt should be forward-looking too.

Capital Structure — Current or Target?

CURRENT MARKET MIX

Use market weights

- Useful when current leverage is representative and stable.

PEER / TARGET MIX

Align with target leverage

- Useful when leverage is moving toward a sustainable industry level.

WARNING

Justify your choice

- Do not use book equity weights for WACC without justification.

Capital structure should match expected financing policy over the valuation horizon.

Worked Example — Assemble WACC

COST OF EQUITY Estimate using CAPM

- R_f : 2.5%
- β : 1.10
- ERP: 5.0%
- K_e : 8.0%

COST OF DEBT Use current credit risk

- Pre-tax K_d : 4.5%
- Tax rate: 25%
- After-tax K_d : 3.375%

TARGET WEIGHTS Use long-run mix

- Equity: 75%
- Debt: 25%

RESULT Compute WACC

- $WACC = 75\% \times 8.0\% + 25\% \times 3.375\% = 6.84\%$

The arithmetic is trivial; the credibility depends on the inputs.

Exercise 1 — Forecast Revenue from Economic Drivers

CASE

A listed medical-device company has 2026 revenue of €2,400m. 60% comes from a mature segment and 40% from a growth segment. Mature-market growth is expected at 2.5%; the company may gain 0.5 percentage points of share. The growth segment market is expected to grow 9%, but competition may reduce the company's share contribution by 1 percentage point. Management guides to “mid-single-digit” group growth.

YOUR TASK

1. Build a 2027 segment-based revenue growth forecast.
2. Calculate implied group revenue growth.
3. Reconcile your estimate with management guidance.
4. Identify three pieces of evidence you would collect before using this assumption in the DCF.
5. State a bear and bull driver — not merely $\pm 1\%$.

Exercise 2 — Margin, ROIC and Reinvestment Consistency

CASE

A consumer company has 2026 revenue of €5,000m, EBIT margin 12%, tax rate 25%, invested capital €3,000m and ROIC 15%. You forecast 6% annual NOPAT growth. Management expects margin to reach 14% in three years while CapEx remains close to D&A and NWC remains flat.

YOUR TASK

1. Estimate the reinvestment rate implied by 6% growth if ROIC remains 15%.
2. Is “CapEx $\approx$ D&A and Δ NWC $\approx$ 0” consistent with that growth?
3. What must be true for margin to rise from 12% to 14%?
4. Give two reasons ROIC may converge downward.
5. Rewrite the forecast assumptions so they tell one coherent economic story.

Exercise 3 — Beta and WACC

CASE

Peer A: β_L 1.20, D/E 30%.

Peer B: β_L 1.05, D/E 20%.

Peer C: β_L 1.35, D/E 50%.

Tax rate = 25%.

Your company's target D/E = 35%.

Assume

R_f = 2.5%,

ERP = 5.0%,

pre-tax cost of debt = 4.5%.

YOUR TASK

1. Unlever the three peer betas.
2. Use the median unlevered beta.
3. Relever at target D/E = 35%.
4. Estimate cost of equity.
5. Convert D/E into debt and equity weights and estimate WACC.
6. Explain why book-value equity weights would be inappropriate.

Exercise 4 — Reinvestment and Working Capital

CASE

Revenue = €4,000m;

NOPAT = €420m.

CapEx = €260m;

D&A = €210m;

R&D expense = €90m.

Net acquisition spending = €140m.

Non-cash NWC falls from -€360m to -€430m.

The company has historically grown partly through acquisitions.

TASK 1

Calculate simple reinvestment using only net CapEx + Δ NWC.

TASK 2

Calculate broader economic reinvestment if R&D and acquisitions are treated as investment.

TASK 3

Interpret the negative NWC change. Is the cash release automatically sustainable?

TASK 4

What evidence would you need before forecasting future reinvestment?

A cash release from working capital can increase FCFF today without representing a sustainable source of long-run value.